北京邮电大学 本科毕业设计（论文）初期进度报告

Project Early-term Progress Report

学院 **School**	International School	**专业** **Programme**	**Telecommunications Engineering with Management**		
姓 **Family name**	Tang	**名** **First Name**	Ziming		
BUPT 学号 **BUPT number**	2022213090	**QM 学号** **QM number**	221165430	**班级** **Class**	2022215103
论文题目 **Project Title**	Tracking progress of Final year projects for students				

Literature Review

1. Github API Integration

A critical requirement of this project is enabling students to manage and track their code versions within the centralized application. Traditional project management systems often lack integrated version control, which makes it difficult for supervisors to monitor iterative progress. The GitHub REST API provides a robust solution by exposing standardized endpoints for repositories, commits, branches, and user authentication.

GitHub. GitHub REST API documentation [Internet]. GitHub Docs; 2022 [cited 2026 Jan 11]. Available from: https://docs.github.com/en/rest?apiVersion=2022-11-28

2. Microsoft Teams API

Effective communication between supervisors and students is central to project success. Traditional systems often rely on email or fragmented messaging, which can slow down feedback loops. The Microsoft Teams API provides standardized endpoints for chat, calls, and collaborative features, enabling seamless integration of real-time communication into the project tracking application.

Microsoft. Build Dashboard with Widget & Graph API - Teams [Internet]. Microsoft Learn; 2023 [cited 2026 Jan 11]. Available from: https://learn.microsoft.com/en-us/microsoftteams/platform/tabs/how-to/build-a-dashboard-tab-app

3. Alibaba Cloud OSS API

File submission and storage are fundamental requirements of the project. Students must upload reports, datasets, and code artifacts, while supervisors need reliable access to these files. The Alibaba Cloud Object Storage Service (OSS) provides scalable, secure, and cost-effective storage solutions, with APIs that support upload, download, and metadata management.

Alibaba Cloud. Object Storage OSS product documentation [Internet]. Alibaba Cloud Docs; 2023 [cited 2026 Jan 11]. Available from: https://www.alibabacloud.com/help/zh/oss

4. System architecture and design principles

A layered architecture (UI, application, data, integration) reduces coupling and supports extensibility for features like progress tracking, version links, storage, and chat. Modeling user flows with UML (use case, sequence, component) and data flow diagrams clarifies requirements and interfaces, enabling consistent implementation across Electron (frontend), Spring Boot (backend), and REST APIs. RESTful design conventions (resources, verbs, pagination, rate limits) are essential to integrate external services (GitHub, Teams) while keeping the internal API maintainable.

1) Fielding RT. Architectural Styles and the Design of Network-based Software Architectures [Internet]. University of California, Irvine; 2000 [cited 2026 Jan 11]. Available from: https://www.ics.uci.edu/~fielding/pubs/dissertation/fielding_dissertation.pdf

2) Richardson L, Ruby S. RESTful Web Services [Internet]. O′Reilly Media; 2007 [cited 2026 Jan 11]. Available from: https://www.oreilly.com/library/view/restful-web-services/9780596529260/

5. Desktop application shell and frontend

Electron enables a cross-platform desktop shell that can host a modern web UI, simplifying deployment while retaining native capabilities. A component-based frontend (React or Vue) supports reusable views for dashboards, student pages, uploads, and blog features. Adopting accessibility and usability heuristics ensures the interfaces for supervisors and students are intuitive, consistent, and inclusive.

OpenJS Foundation. Electron documentation [Internet]. Electron; 2025 [cited 2026 Jan 11]. Available from: https://www.electronjs.org/docs/latest

6. Backend framework, database, and caching

Spring Boot streamlines backend service development with production-ready configurations and security integrations. MySQL provides relational consistency for user accounts, progress logs, and blog metadata. Redis offers low-latency caching for frequently accessed views (e.g., supervisor dashboards) and session management, improving responsiveness under concurrent load.

1) Pivotal/VMware. Spring Boot reference documentation [Internet]. Spring; 2025 [cited 2026 Jan 11]. Available from: https://docs.spring.io/spring-boot/docs/current/reference/html/
2) Oracle. MySQL 8.0 reference manual [Internet]. Oracle; 2025 [cited 2026 Jan 11]. Available from: https://dev.mysql.com/doc/refman/8.0/en/
3) **Redis Ltd. Redis documentation [Internet]. Redis; 2025 [cited 2026 Jan 11]. Available from: https://redis.io/docs/**

7. **Testing and quality assurance**

Unit tests validate services and controllers; integration tests verify frontend–backend–database interactions; system tests assess end-to-end scenarios like login, upload, and version view. Automated testing with JUnit (backend), Cypress (frontend), and API contract tests improves reliability and supports mid-term assessment. CI pipelines ensure repeatable builds and test runs.

JUnit Team. JUnit 5 user guide [Internet]. JUnit; 2025 [cited 2026 Jan 11]. Available from: https://junit.org/junit5/docs/current/user-guide/

8. Blog and peer engagement features

Lightweight social features (comments, seen, likes) can be designed using proven interaction patterns to support peer learning and reflective practice. Moderation and audit logs ensure accountability. Aligning with usability and accessibility standards keeps the feature inclusive.

Preece J, Rogers Y, Sharp H. Interaction Design: Beyond Human-Computer Interaction [Internet]. 5th ed. Wiley; 2019 [cited 2026 Jan 11]. Available from: https://www.wiley.com/en-us/Interaction+Design%3A+Beyond+Human+Computer+Interaction%2C+5th+Edition-p-9781119547259

9. Logging, observability, and performance

Structured logging and metrics (e.g., Spring Boot Actuator) enable monitoring of authentication, uploads, and API calls. Application performance can be profiled to ensure responsive dashboards and student pages, especially under load. Observability supports auditability, which is relevant to ethical accountability.

Grafana Labs. Grafana documentation [Internet]. Grafana; 2025 [cited 2026 Jan 11]. Available from: https://grafana.com/docs/

Progress Summary

是否符合进度？On schedule as per GANTT chart?

[YES/NO – if answer NO, give reasons and a plan to catch up. DELETE THIS LINE and WHITE SPACES]

YES, Task1 all completed, yet Task2 is still in progress.

下一步 Next steps:

[Focus on the next immediate steps, i.e. until the next report – DELETE THIS LINE and WHITE SPACES]

Proceed Task2 and look for more solutions.